

first light

digitization and programming · setup and first steps · prof. dr. nicolas meseth

- 1 your device
- 2 your assistant
- 3 getting ready
- 4 your first program
- 5 how our workshops work

part 1

your device



image: ai-generated from photos of the real device (gpt-image-2.5-sunburst)

your device

the eye

a colour sensor, on the left



your device

your device

the eye

a colour sensor, on the left

the lamp

an RGB LED, on the right



your device

your device

the eye

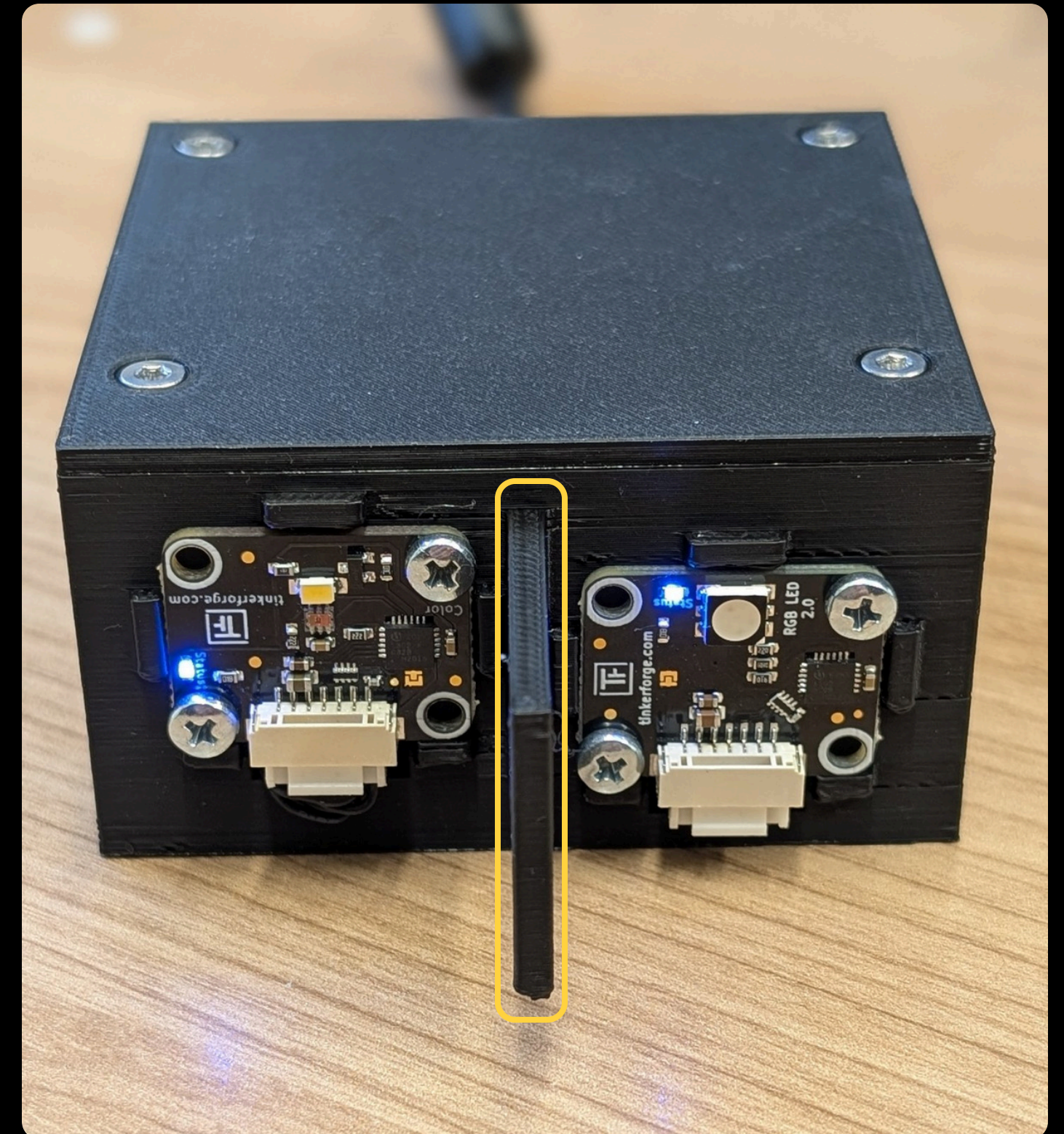
a colour sensor, on the left

the lamp

an RGB LED, on the right

the wall

so the device does not see its own lamp



your device

the eye

a colour sensor, on the left

the lamp

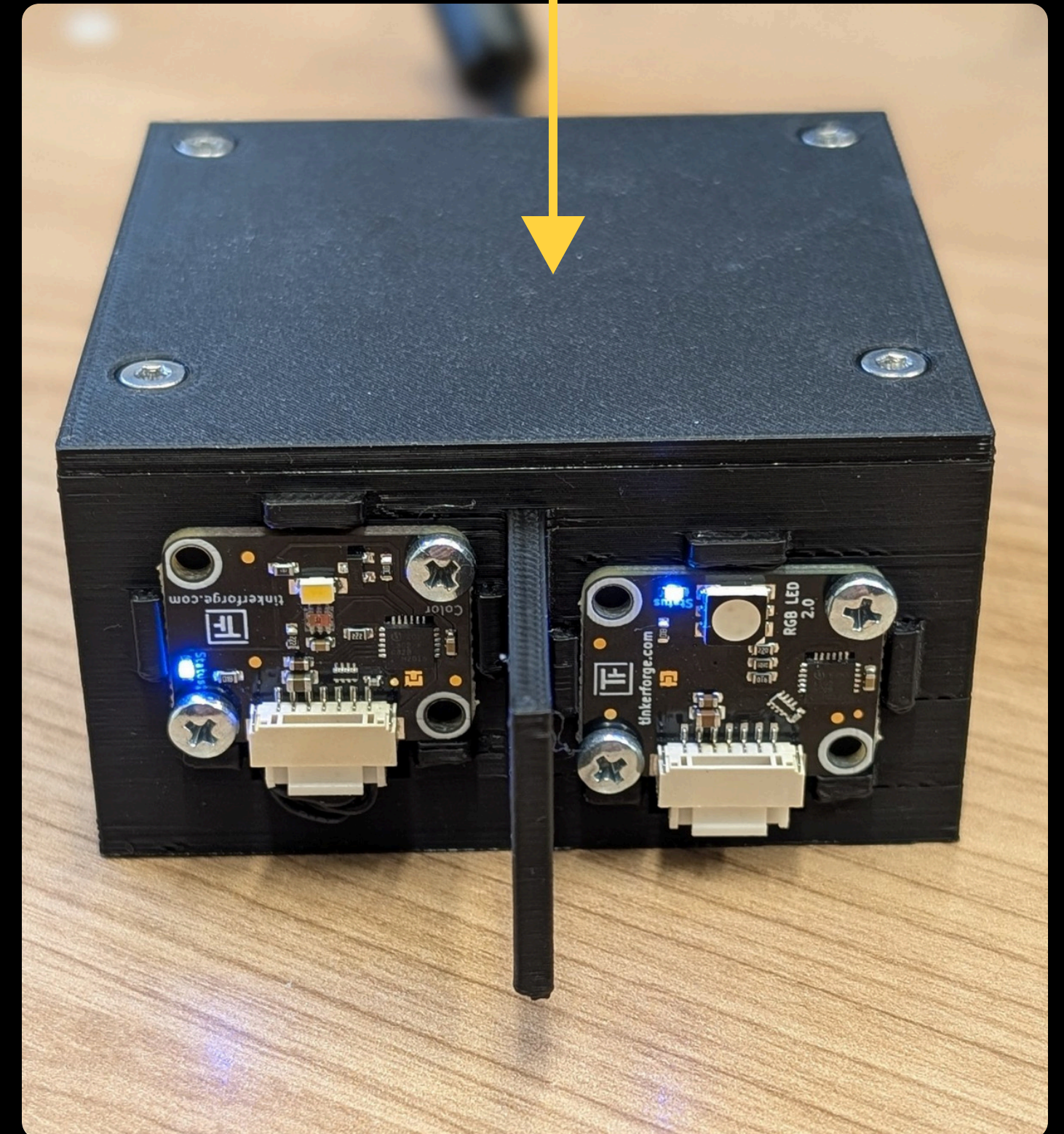
an RGB LED, on the right

the wall

so the device does not see its own lamp

inside

a Master Brick, connected to your laptop by USB



 yours from next monday

two devices per team, one for each of you.
you get them next monday, and keep them until january.
find your partner by then. today, your laptop gets ready.

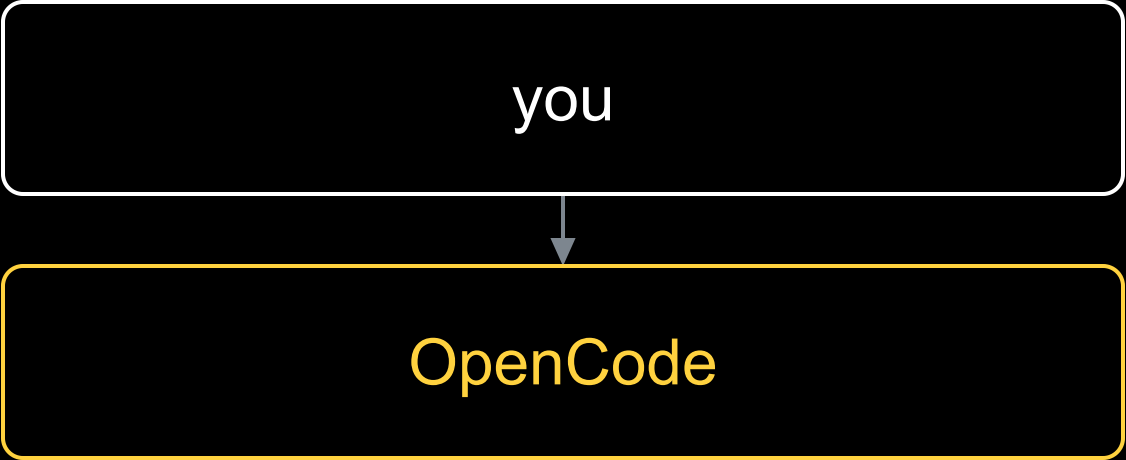
from your idea to the light



decide what should happen, and write code



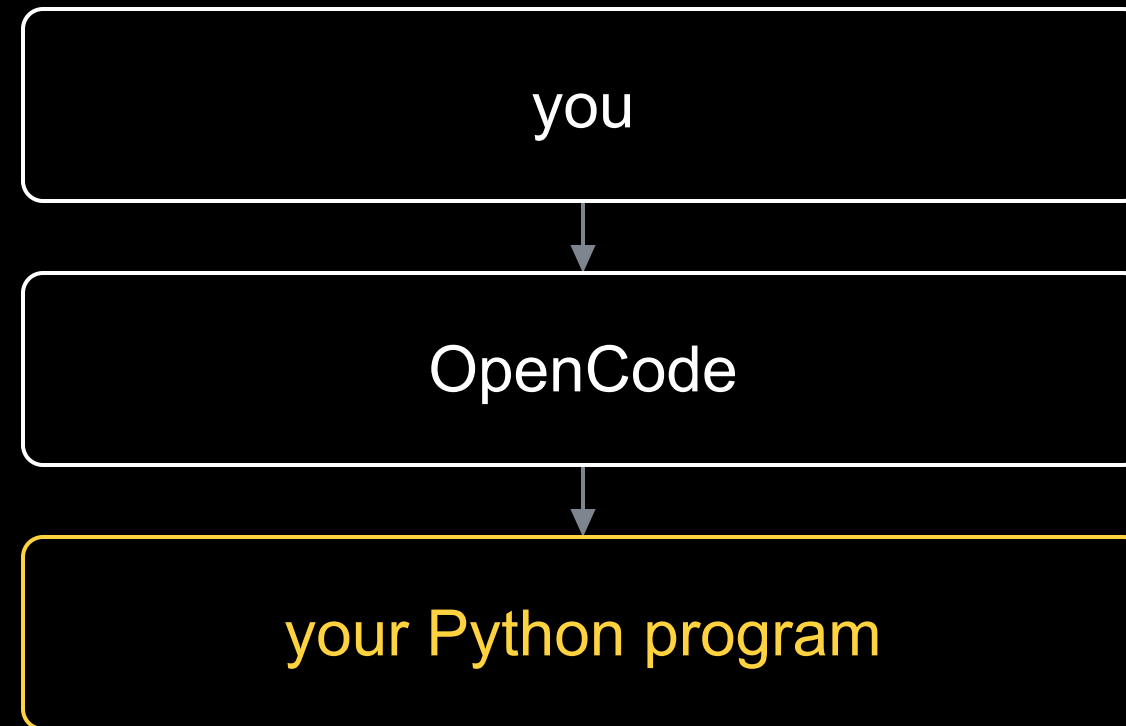
from your idea to the light



decide what should happen, and write code

your assistant helps you write and understand it

from your idea to the light

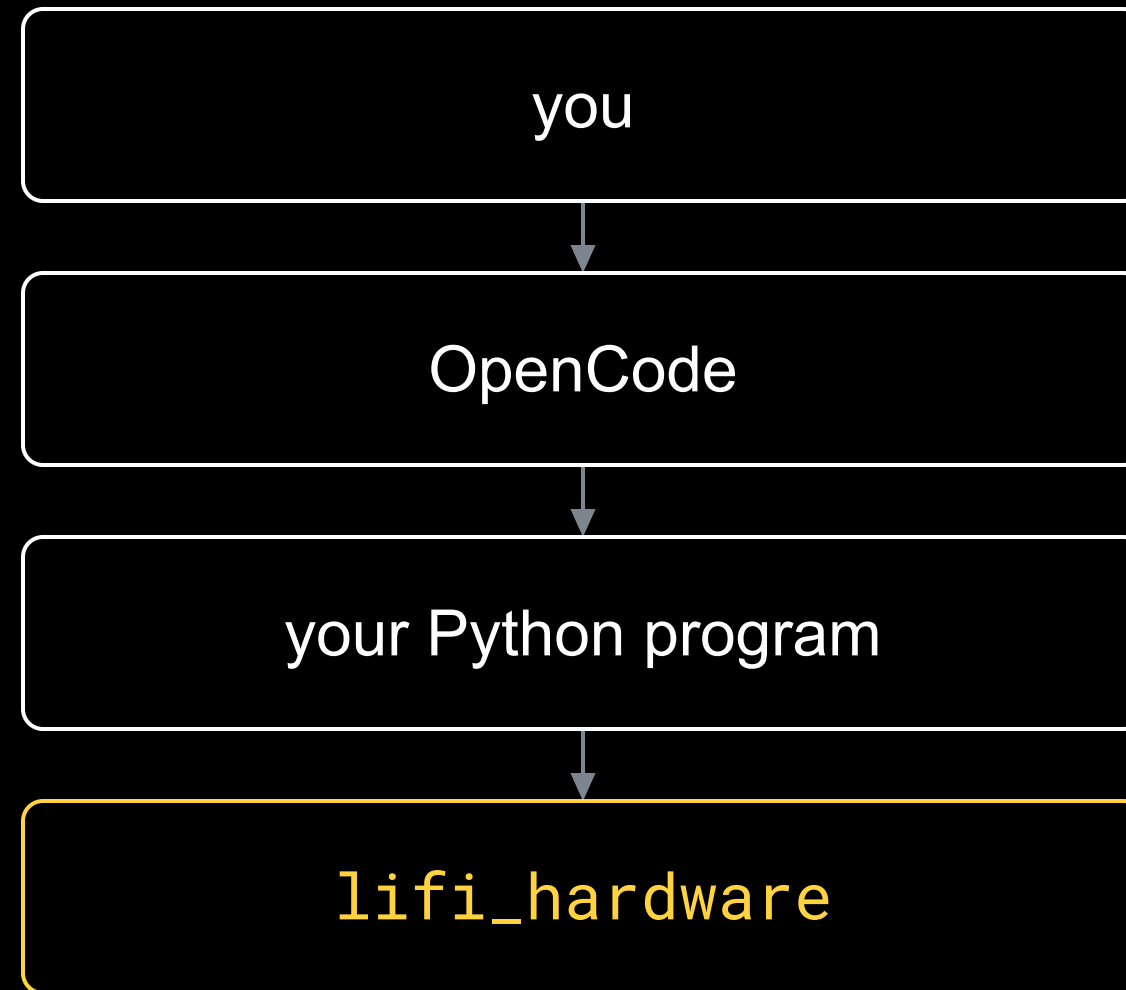


decide what should happen, and write code

your assistant helps you write and understand it

the code you write together, in my-code

from your idea to the light



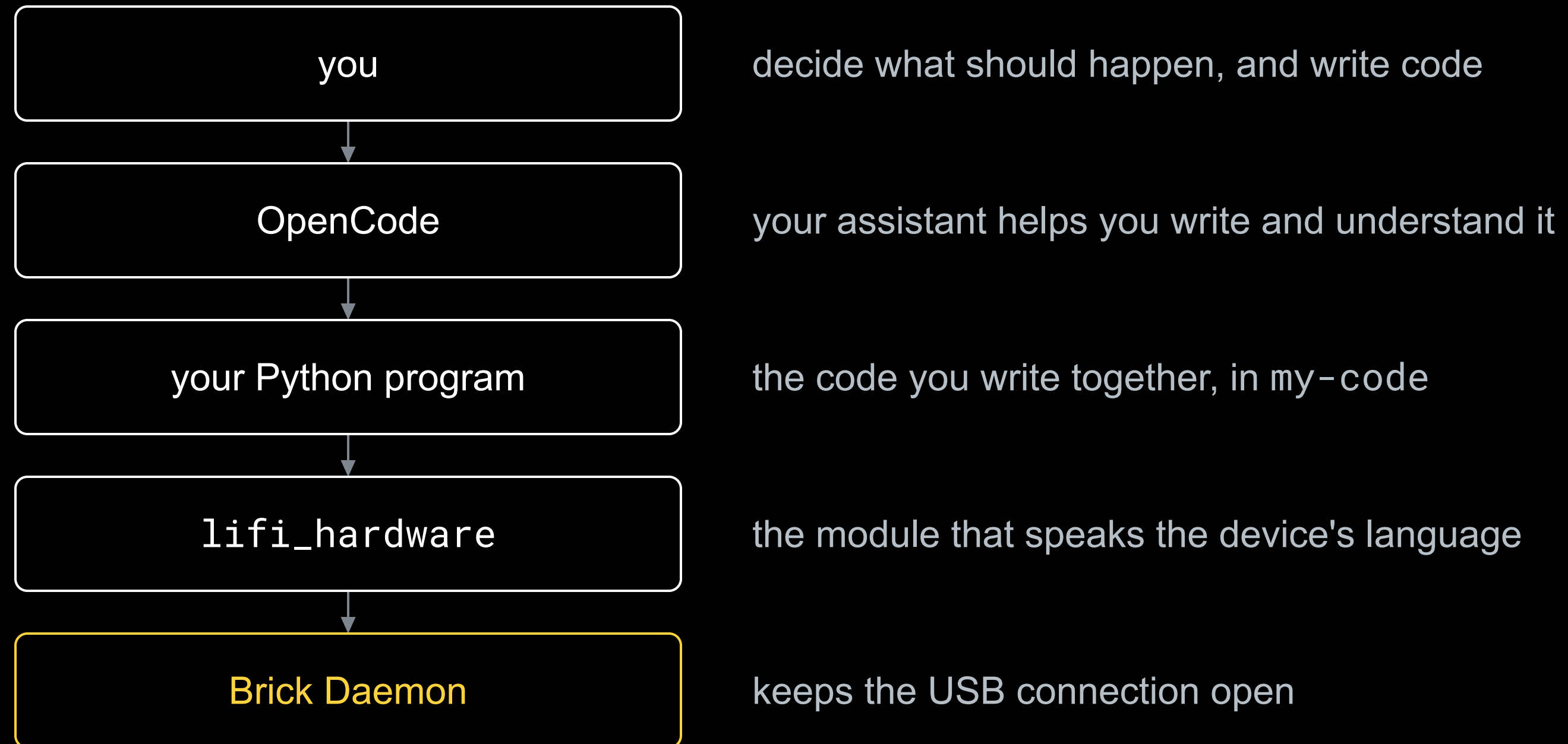
decide what should happen, and write code

your assistant helps you write and understand it

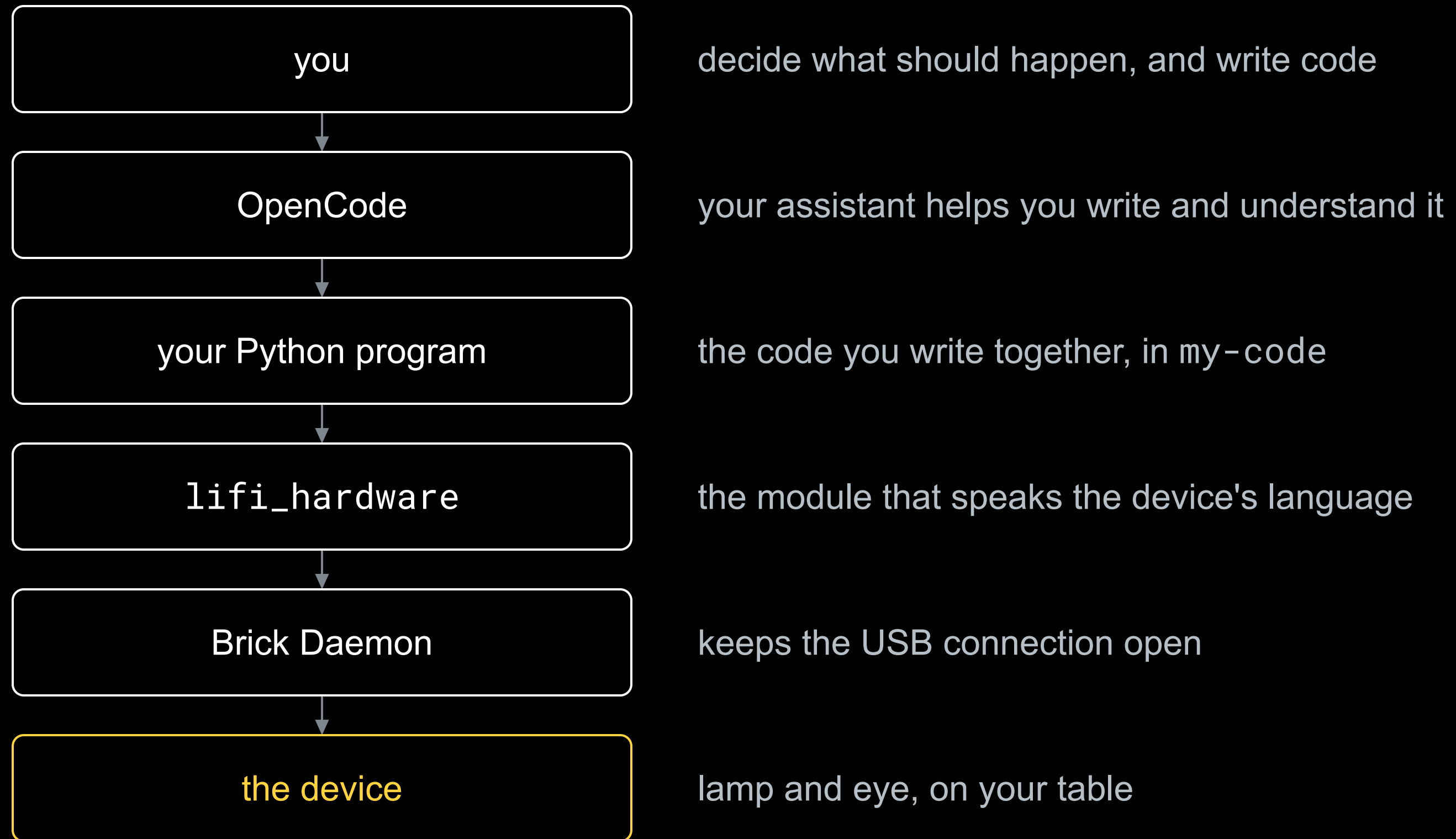
the code you write together, in my-code

the module that speaks the device's language

from your idea to the light



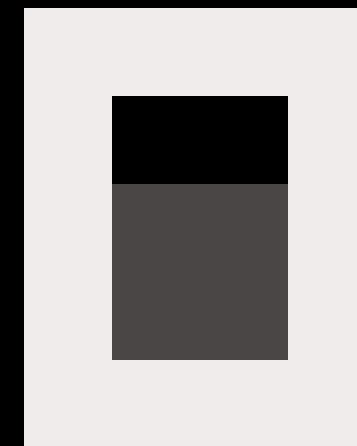
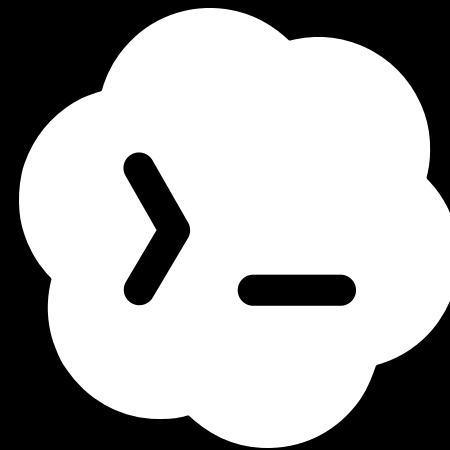
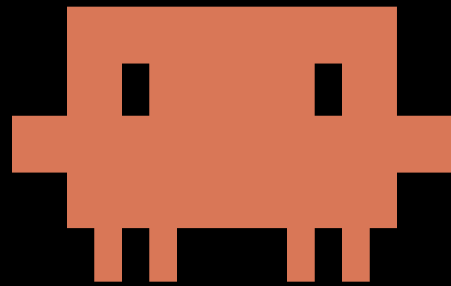
from your idea to the light



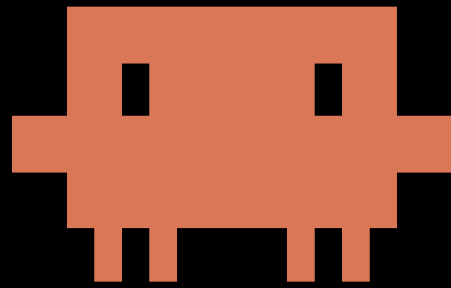
part 2

your assistant

do you know who these are?

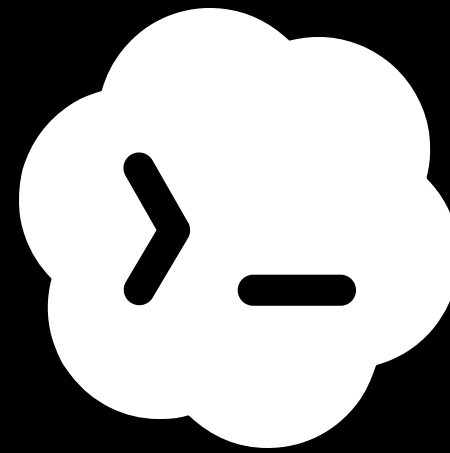


do you know who these are?



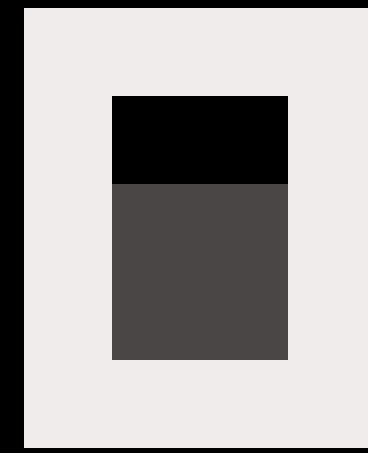
Claude Code

Anthropic



Codex

OpenAI



OpenCode

open source

you know this already

a chat, like ChatGPT

one question, one answer

an agent, like OpenCode

one task, until it is done

you know this already

a chat, like ChatGPT

one question, one answer

you paste in what it needs

an agent, like OpenCode

one task, until it is done

it reads your files itself

you know this already

a chat, like ChatGPT

one question, one answer

you paste in what it needs

you copy its code into a file

an agent, like OpenCode

one task, until it is done

it reads your files itself

it writes the file itself

you know this already

a chat, like ChatGPT

one question, one answer

you paste in what it needs

you copy its code into a file

you run it and report back

an agent, like OpenCode

one task, until it is done

it reads your files itself

it writes the file itself

it runs it and reads the result

you know this already

a chat, like ChatGPT

one question, one answer

you paste in what it needs

you copy its code into a file

you run it and report back

you decide the next step

an agent, like OpenCode

one task, until it is done

it reads your files itself

it writes the file itself

it runs it and reads the result

it tries again, until it works

you know this already

a chat, like ChatGPT

one question, one answer

you paste in what it needs

you copy its code into a file

you run it and report back

you decide the next step

an agent, like OpenCode

one task, until it is done

it reads your files itself

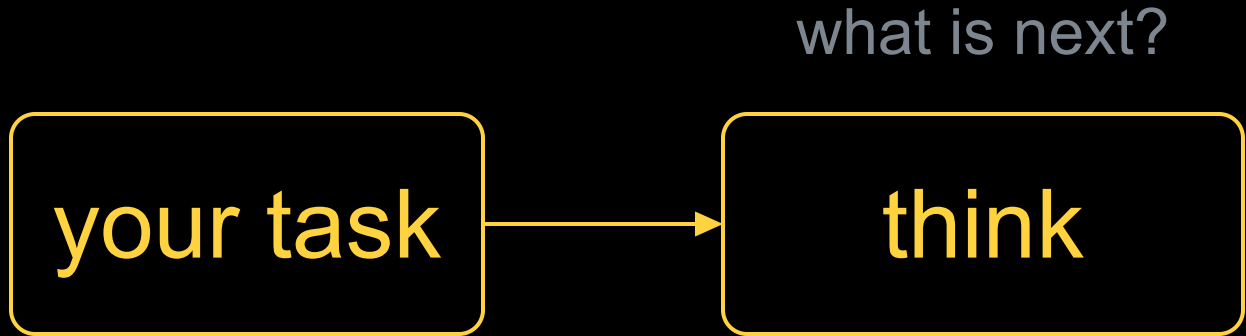
it writes the file itself

it runs it and reads the result

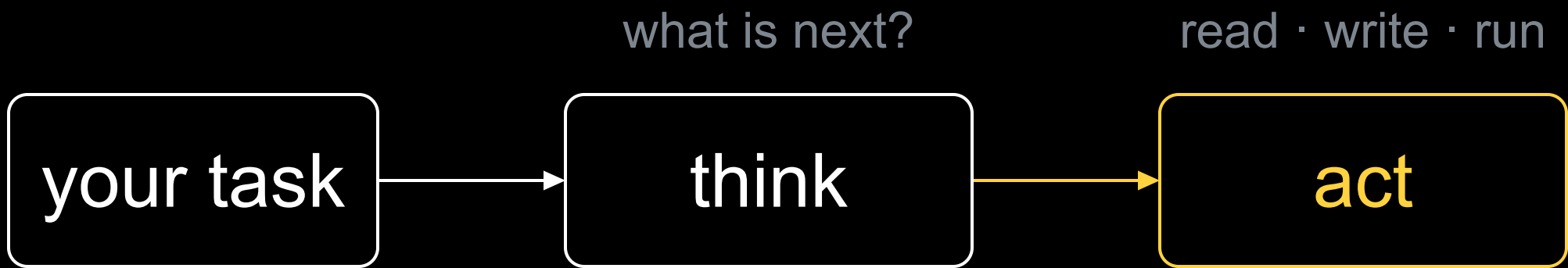
it tries again, until it works

the difference: it works in a loop, right where the work is.

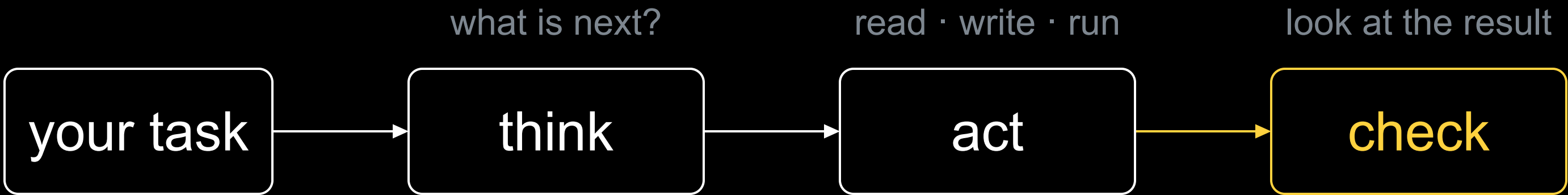
the agent loop



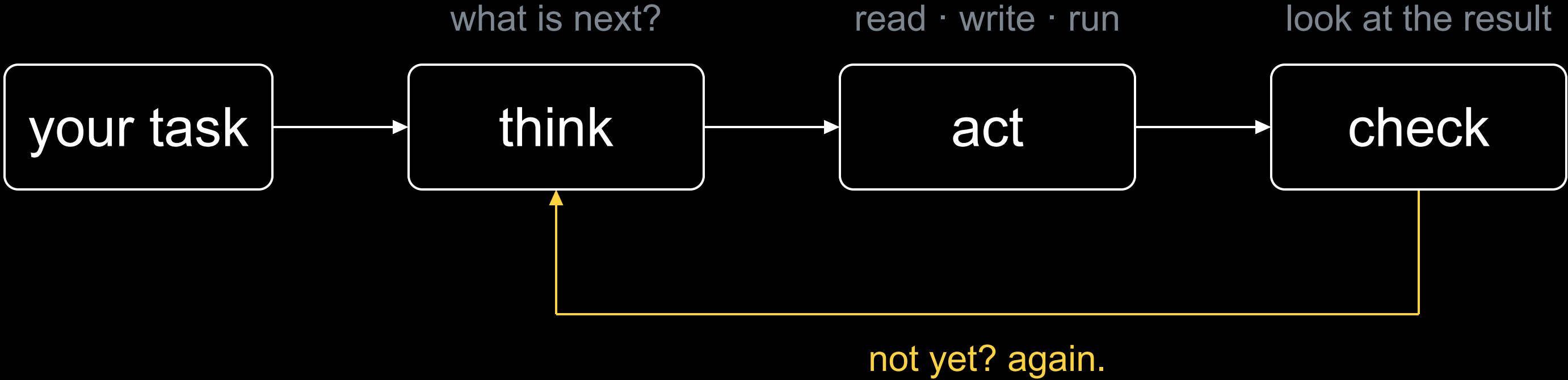
the agent loop



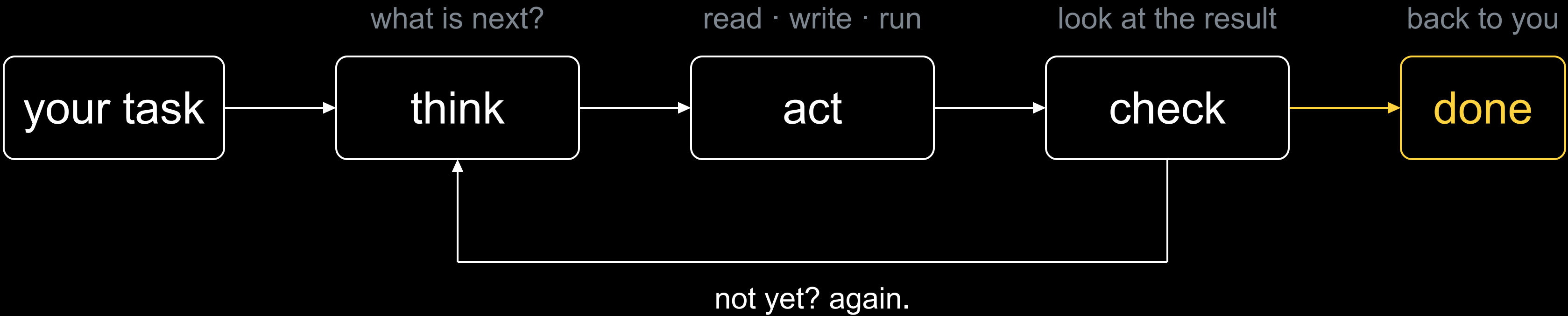
the agent loop



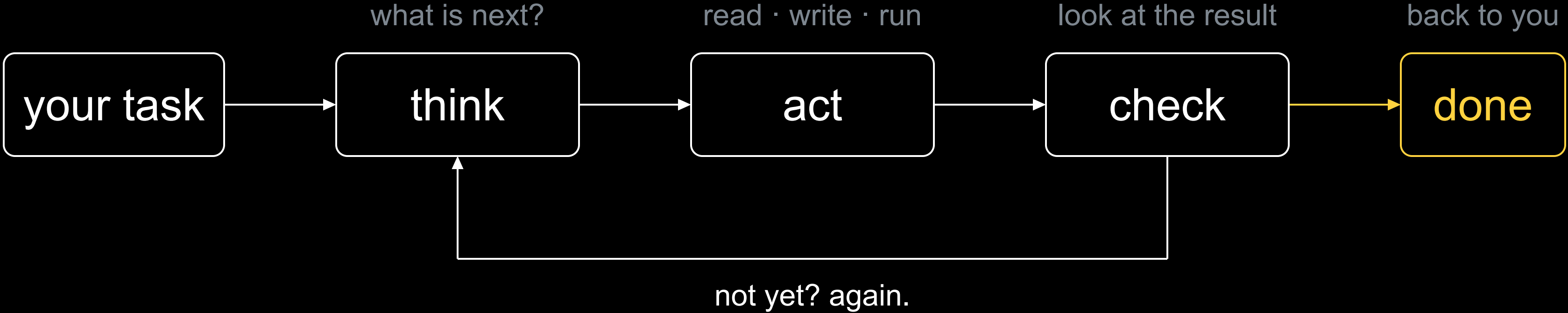
the agent loop



the agent loop



the agent loop



you give the task. you check the result.



your assistant, set up for this course.



works on your
laptop

in your course folder, with
your files.



uses tools

reads and writes files, runs
programs and commands.



knows the course

the website, the lecture
notes, the module
lifi_hardware.



it cannot see your
device

the lamp has the last word.

the program and the model

on your laptop

opencode

your files

terminal

the program and the model

language model

☁ a server, somewhere else

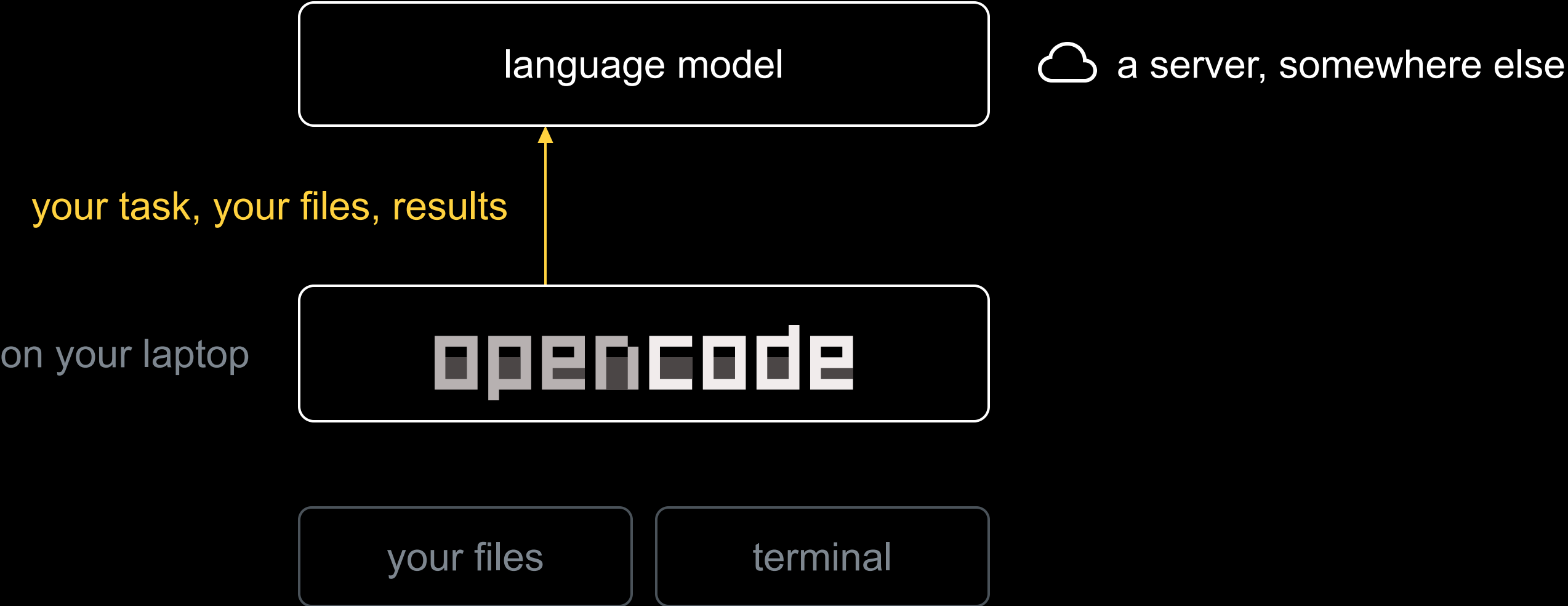
on your laptop

opencode

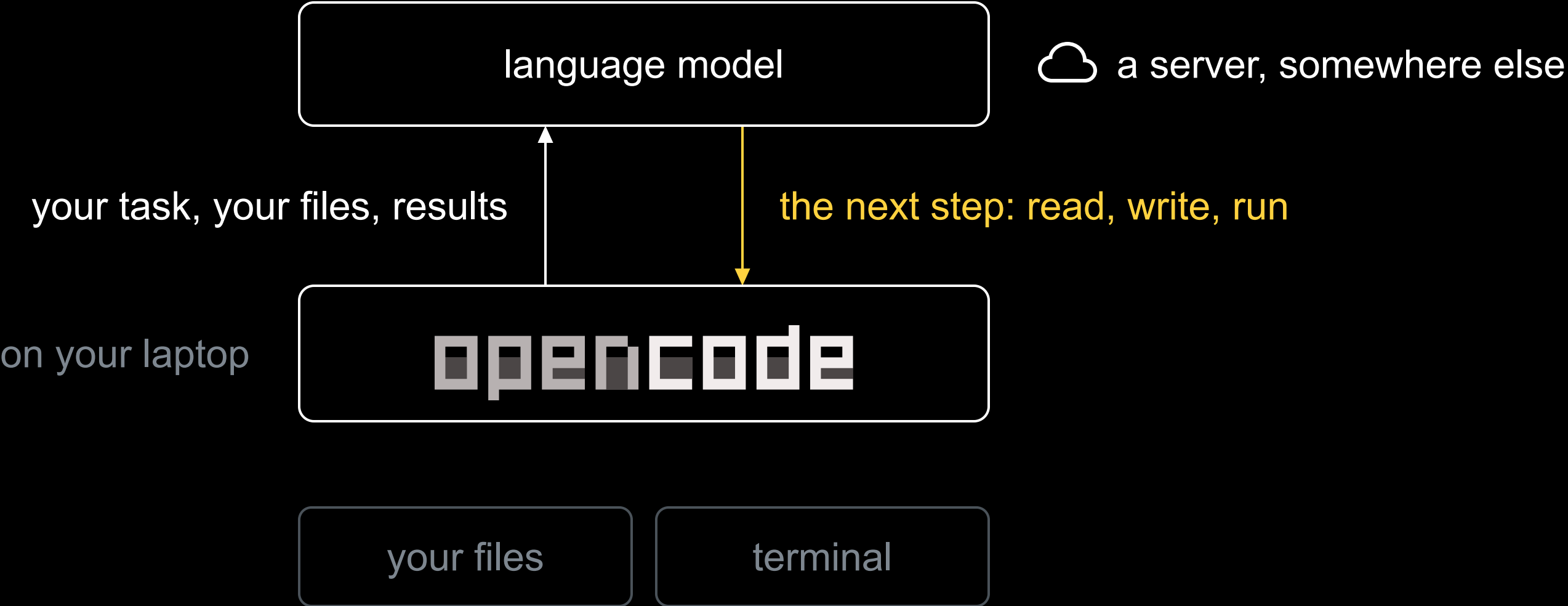
your files

terminal

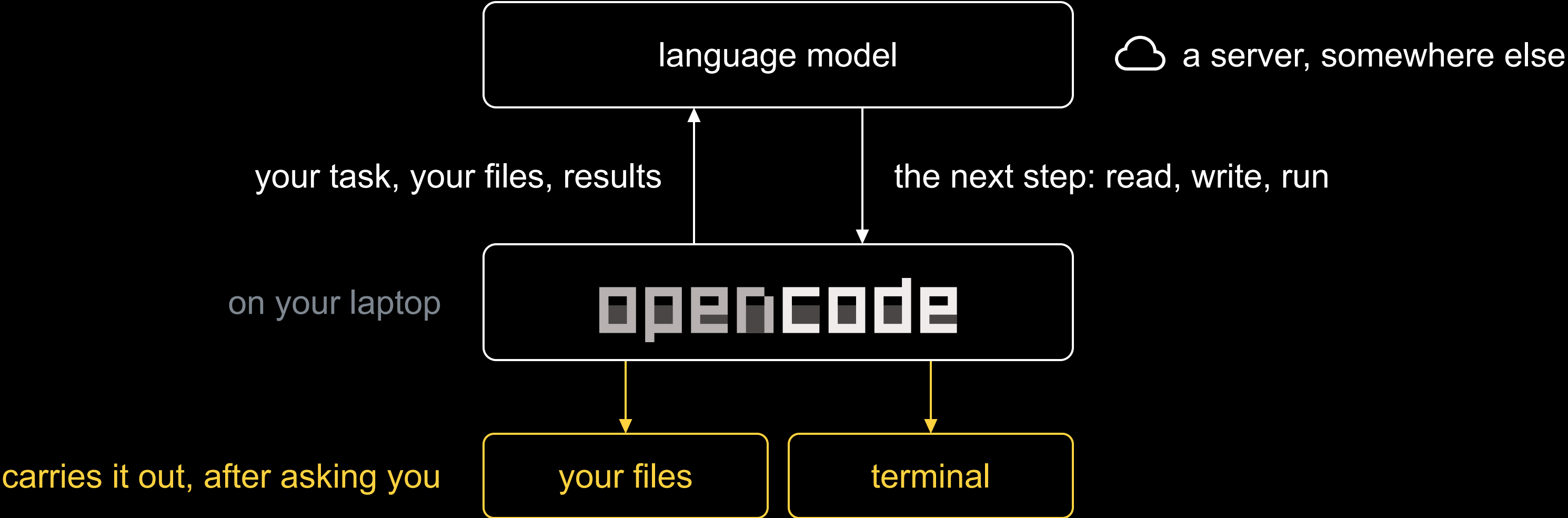
the program and the model



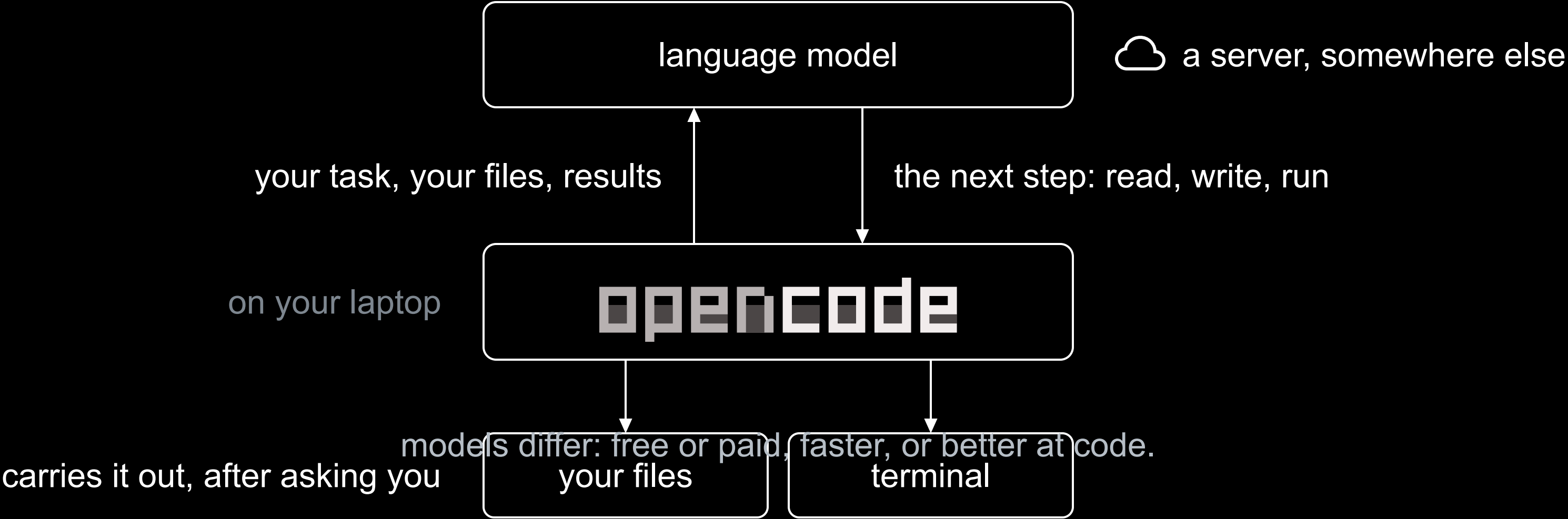
the program and the model



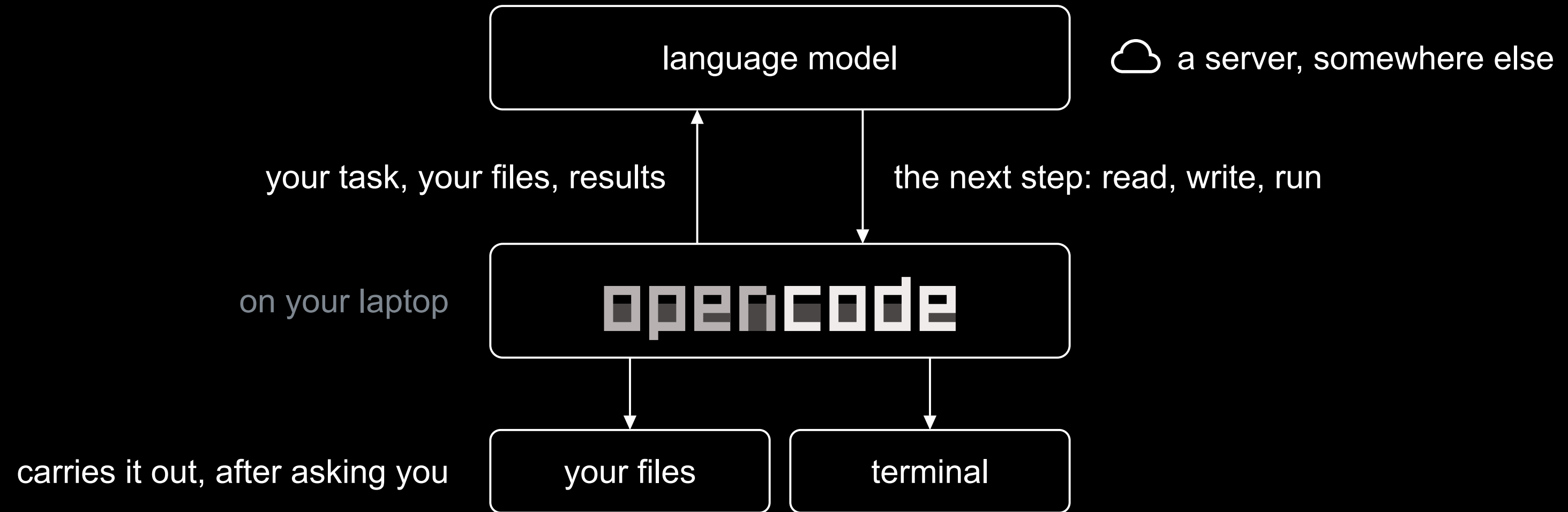
the program and the model



the program and the model



the program and the model



models differ: free or paid, faster, or better at code.

what you type travels to that server.

your model



free models first

OpenCode comes with free models. they work right away, without a key.



a key, if needed

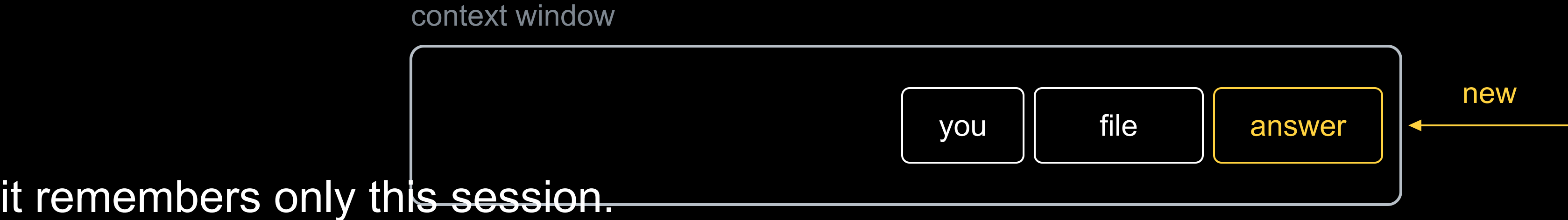
for a stronger model, a key may follow later. you will hear about it.



a key is only for OpenCode

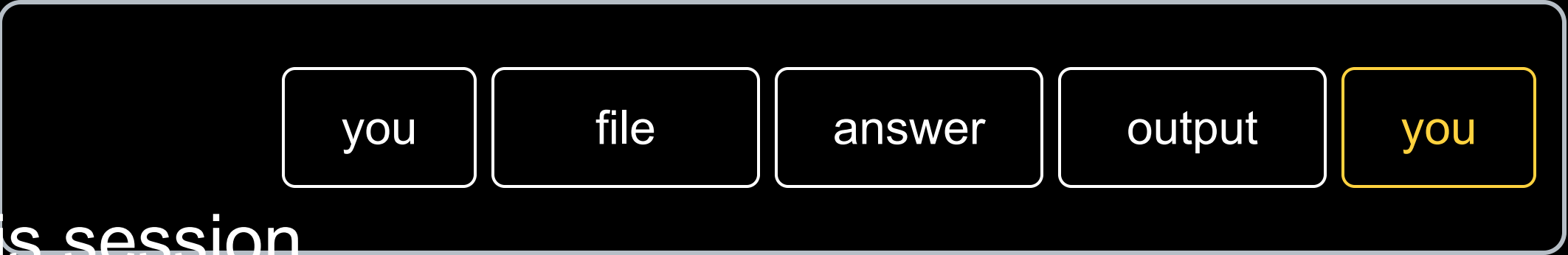
never into a chat, a file or a screenshot. whoever has a key works at its owner's expense.

its memory: the context window



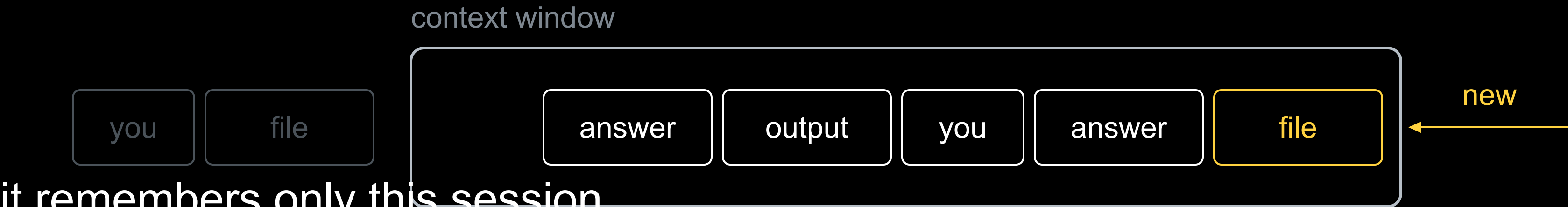
its memory: the context window

context window



it remembers only this session.
the window has a size, counted in tokens.

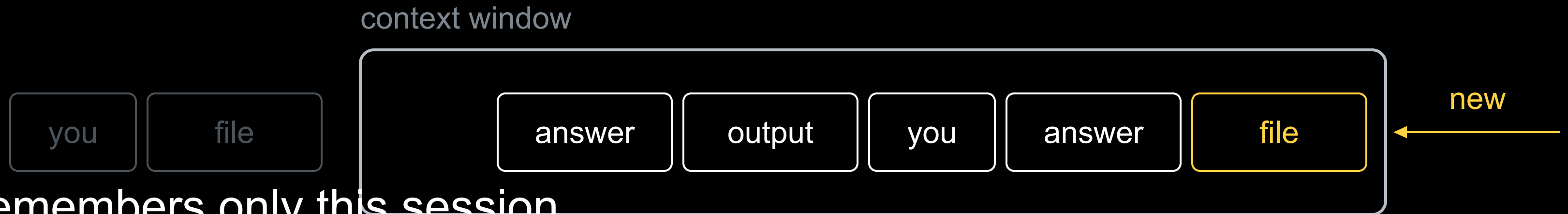
its memory: the context window



the window has a size, counted in tokens.

when it is full, it compacts: old parts shrink to a summary.

its memory: the context window



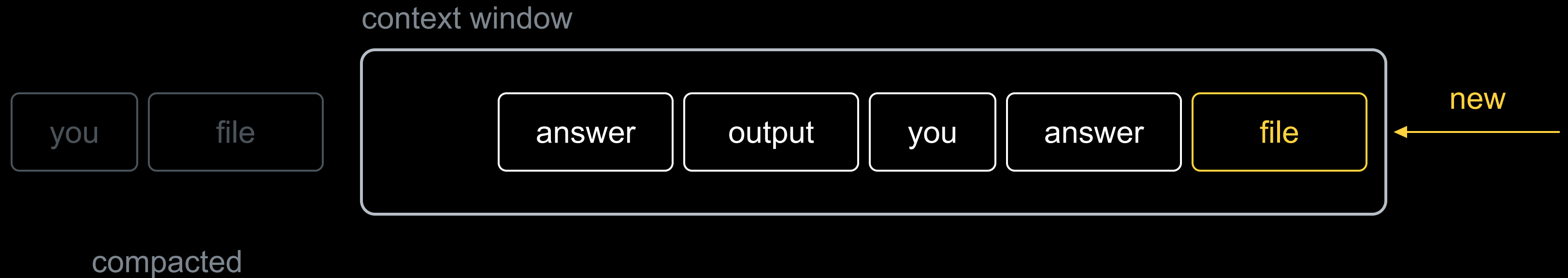
it remembers only this session.

the window has a size, counted in tokens.

when it is full, it compacts: old parts shrink to a summary.

new task, new session.

its memory: the context window



it remembers only this session.

the window has a size, counted in tokens.

when it is full, it compacts: old parts shrink to a summary.

new task, new session.

what should last goes into a file it reads at every start, like its course instructions.

✓ careful, not scared

it asks first

changes and commands wait for your ok. read,
then allow.

 careful, not scared

it asks first

changes and commands wait for your ok. read,
then allow.

it works in your course folder

keep private files out of it.

✓ careful, not scared

it asks first

changes and commands wait for your ok. read,
then allow.

it works in your course folder

keep private files out of it.

it can be wrong

and it sounds just as sure. run it, measure it,
check it.

✓ careful, not scared

it asks first

changes and commands wait for your ok. read, then allow.

it works in your course folder

keep private files out of it.

it can be wrong

and it sounds just as sure. run it, measure it, check it.

it can make mistakes

one ok too quick, and a file is gone. keep a backup of what matters.

✓ careful, not scared

it asks first

changes and commands wait for your ok. read, then allow.

it works in your course folder

keep private files out of it.

it can be wrong

and it sounds just as sure. run it, measure it, check it.

it can make mistakes

one ok too quick, and a file is gone. keep a backup of what matters.

you can stop it

any time, with the stop button.

✓ careful, not scared

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changes and commands wait for your ok. read, then allow.

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and it sounds just as sure. run it, measure it, check it.

it can make mistakes

one ok too quick, and a file is gone. keep a backup of what matters.

you can stop it

any time, with the stop button.

your chat leaves your laptop

no passwords, no keys, no personal data in the chat.

✓ careful, not scared

it asks first

changes and commands wait for your ok. read, then allow.

it works in your course folder

keep private files out of it.

it can be wrong

and it sounds just as sure. run it, measure it, check it.

it can make mistakes

one ok too quick, and a file is gone. keep a backup of what matters.

you can stop it

any time, with the stop button.

your chat leaves your laptop

no passwords, no keys, no personal data in the chat.

you decide what it may do.

part 3

getting ready

part 1, by hand

1 install OpenCode Desktop.

step by step: docs.lifi-project.de/software

part 1, by hand

- 1 install OpenCode Desktop.
- 2 download the course folder as a zip and unpack it.

step by step: docs.lifi-project.de/software

part 1, by hand

- 1 install OpenCode Desktop.
- 2 download the course folder as a zip and unpack it.
- 3 open the folder in OpenCode and start a session.

step by step: docs.lifi-project.de/software

part 2, your assistant takes over

/onboarding



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.

it installs Python, lifi_hardware, VS Code, Brick Daemon and Brick Viewer.



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.

it installs Python, `lifi_hardware`, VS Code, Brick Daemon and Brick Viewer.

it checks every step before the next one.



part 2, your assistant takes over

/onboarding

it asks you three questions.

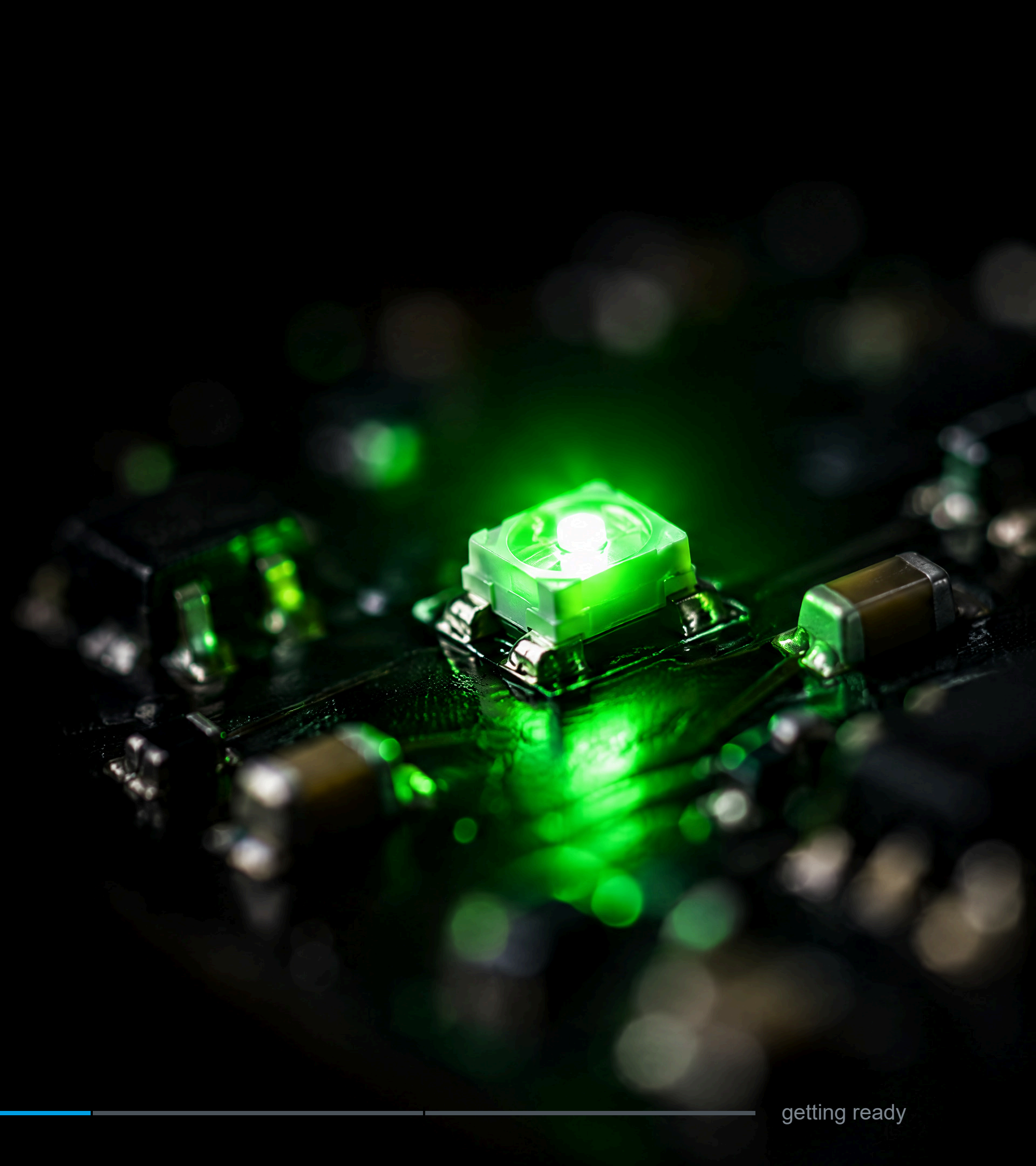
it installs Python, lifi_hardware, VS Code, Brick Daemon and Brick Viewer.

it checks every step before the next one.

no device yet? it skips the device test.

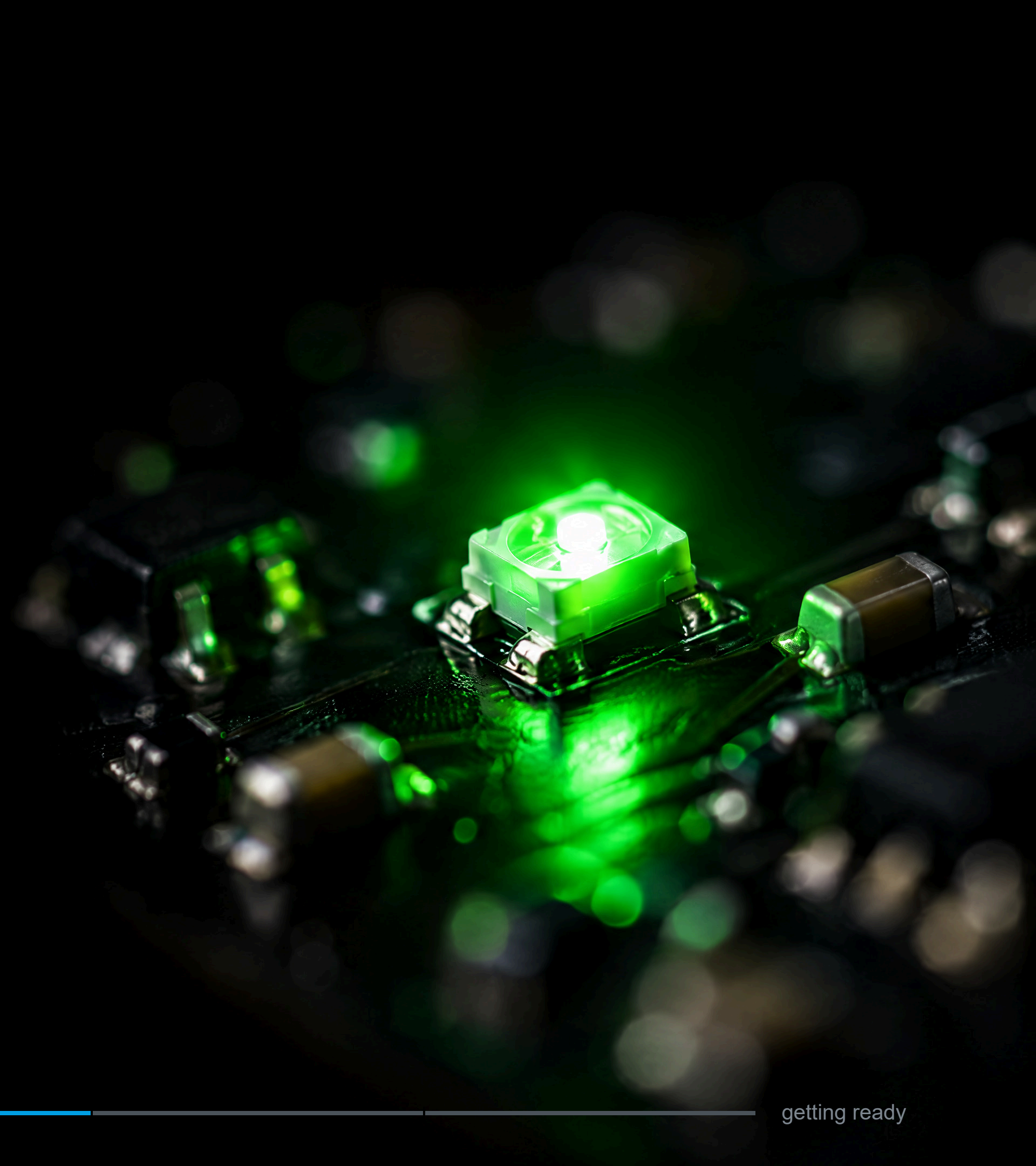


your laptop is ready for
your device.



your laptop is ready for your device.

next monday, this LED lights up green.



part 4

your first program

a program is a file

in VS Code, create a new file in my-code: hello.py

a program is a file

in VS Code, create a new file in my-code: hello.py
the ending .py says: this text is Python.

a program is a file

in VS Code, create a new file in my-code: hello.py
the ending .py says: this text is Python.

a program is plain text,
until Python runs it.

your first line

```
print("Hello, world!")
```

your first line

```
print("Hello, world!")
```

save it. then open a terminal (terminal › new terminal) and type:

```
cd my-code  
python hello.py
```

on a mac: python3 instead of python.

your first line

```
print("Hello, world!")
```

save it. then open a terminal (terminal › new terminal) and type:

```
cd my-code  
python hello.py
```

```
Hello, world!
```

on a mac: python3 instead of python.

asking and answering

```
name = input("What is your name? ")  
print("Hello, " + name + "!!")
```

asking and answering

```
name = input("What is your name? ")  
print("Hello, " + name + "!!")
```

```
What is your name? Ada  
Hello, Ada!
```


asking and answering

```
name = input("What is your name? ")  
print("Hello, " + name + " !")
```

```
What is your name? Ada  
Hello, Ada!
```

`input()` waits until you press enter, and hands back what you typed.

asking and answering

```
name = input("What is your name? ")  
print("Hello, " + name + " !")
```

```
What is your name? Ada  
Hello, Ada!
```

`input()` waits until you press enter, and hands back what you typed.

`name` is a variable: a name for that value, so you can use it again.

the task

write a timer for the terminal. it asks how many seconds, then counts down, one line per second.

a coloured lamp shows how it stands:
green while it runs, yellow in the last five seconds, red when time is up.



cut it into steps

- 1 get the number of seconds from the user

cut it into steps

- 1 get the number of seconds from the user
- 2 check if the input is valid

cut it into steps

- 1 get the number of seconds from the user
- 2 check if the input is valid
- 3 count down each second

cut it into steps

- 1 get the number of seconds from the user
- 2 check if the input is valid
- 3 count down each second
- 4 show a coloured lamp

cut it into steps

- 1 get the number of seconds from the user
- 2 check if the input is valid
- 3 count down each second
- 4 show a coloured lamp
- 5 decide which colour

cut it into steps

1 get the number of seconds from the user

`input()`

2 check if the input is valid

`while`

3 count down each second

`for`

4 show a coloured lamp

`def`

5 decide which colour

`if / else`

cut it into steps

- | | | |
|---|---|------------------------|
| 1 | get the number of seconds from the user | <code>input()</code> |
| 2 | check if the input is valid | <code>while</code> |
| 3 | count down each second | <code>for</code> |
| 4 | show a coloured lamp | <code>def</code> |
| 5 | decide which colour | <code>if / else</code> |

one step at a time, and run it after each one.

 and your assistant?

ask it like this:

“explain what `input()` hands back.”

 and your assistant?

ask it like this:

“explain what `input()` hands back.”

“read my `timer.py`. is step 2 right?”

 and your assistant?

ask it like this:

“explain what `input()` hands back.”

“read my `timer.py`. is step 2 right?”

“run it with five as input. what goes wrong?”

 and your assistant?

ask it like this:

“explain what `input()` hands back.”

“read my `timer.py`. is step 2 right?”

“run it with five as input. what goes wrong?”

“give me a hint for step 3, not the code.”

 and your assistant?

ask it like this:

“explain what `input()` hands back.”

“read my `timer.py`. is step 2 right?”

“run it with five as input. what goes wrong?”

“give me a hint for step 3, not the code.”

let it explain, not only write.

where we are · next: step 1

- | | | |
|---|---|------------------------|
| 1 | get the number of seconds from the user | <code>input()</code> |
| 2 | check if the input is valid | <code>while</code> |
| 3 | count down each second | <code>for</code> |
| 4 | show a coloured lamp | <code>def</code> |
| 5 | decide which colour | <code>if / else</code> |

step 1 · get the number of seconds

```
text = input("How many seconds? ")  
seconds = int(text)  
print("Timer:", seconds, "seconds")
```

step 1 · get the number of seconds

```
text = input("How many seconds? ")  
seconds = int(text)  
print("Timer:", seconds, "seconds")
```

`input()` always hands back text, even if you type 20.

step 1 · get the number of seconds

```
text = input("How many seconds? ")  
seconds = int(text)  
print("Timer:", seconds, "seconds")
```

`input()` always hands back text, even if you type 20.

`int()` turns the text into a whole number, so you can count with it.

what if someone types five?

How many seconds? five

```
Traceback (most recent call last):  
  File "...\\my-code\\timer.py", line 2, in <module>  
    seconds = int(text)  
ValueError: invalid literal for int() with base 10: 'five'
```

what if someone types five?

How many seconds? five

```
Traceback (most recent call last):  
  File "...\\my-code\\timer.py", line 2, in <module>  
    seconds = int(text)  
ValueError: invalid literal for int() with base 10: 'five'
```

read it from the bottom: the last line says what went wrong.

what if someone types five?

How many seconds? five

Traceback (most recent call last):

File "...\\my-code\\timer.py", line 2, in <module>

seconds = int(text)

ValueError: invalid literal for int() with base 10: 'five'

read it from the bottom: the last line says what went wrong.
above it: where. the file, line 2, and the line itself.

what if someone types five?

How many seconds? five

```
Traceback (most recent call last):  
  File "...\\my-code\\timer.py", line 2, in <module>  
    seconds = int(text)  
ValueError: invalid literal for int() with base 10: 'five'
```

read it from the bottom: the last line says what went wrong.
above it: where. the file, line 2, and the line itself.

stuck? ask your assistant.

it can run your program itself and read the error.

where we are · next: step 2

- | | | |
|---|---|------------------------|
| ✓ | get the number of seconds from the user | <code>input()</code> |
| 2 | check if the input is valid | <code>while</code> |
| 3 | count down each second | <code>for</code> |
| 4 | show a coloured lamp | <code>def</code> |
| 5 | decide which colour | <code>if / else</code> |

step 2 · check if the input is valid

```
text = input("How many seconds? ")  
while not text.isdigit():  
    text = input("Digits only, please: ")  
seconds = int(text)
```

step 2 · check if the input is valid

```
text = input("How many seconds? ")  
while not text.isdigit():  
    text = input("Digits only, please: ")  
seconds = int(text)
```

`text.isdigit()` is true if the text has only digits.

step 2 · check if the input is valid

```
text = input("How many seconds? ")  
while not text.isdigit():  
    text = input("Digits only, please: ")  
seconds = int(text)
```

`text.isdigit()` is true if the text has only digits.

`while` asks again, as long as the answer is not a number.

where we are · next: step 3

✓ get the number of seconds from the user

`input()`

✓ check if the input is valid

`while`

3 count down each second

`for`

4 show a coloured lamp

`def`

5 decide which colour

`if / else`

step 3 · count down each second

```
import time

for remaining in range(seconds, 0, -1):
    print(remaining, "s")
    time.sleep(1)

print("time is up!")
```

step 3 · count down each second

```
import time

for remaining in range(seconds, 0, -1):
    print(remaining, "s")
    time.sleep(1)

print("time is up!")
```

`range(seconds, 0, -1)` counts down from seconds to 1.

step 3 · count down each second

```
import time

for remaining in range(seconds, 0, -1):
    print(remaining, "s")
    time.sleep(1)

print("time is up!")
```

`range(seconds, 0, -1)` counts down from seconds to 1.

the indented lines repeat, once for every number. `time.sleep(1)` waits one second.

where we are · next: step 4

✓ get the number of seconds from the user

`input()`

✓ check if the input is valid

`while`

✓ count down each second

`for`

4 **show a coloured lamp**

`def`

5 decide which colour

`if / else`

step 4 · show a coloured lamp

```
def lamp(r, g, b):  
    on = f"\033[48;2;{r};{g};{b}m"  
    off = "\033[0m"  
    print(on + " " * 12 + off, end=" ")
```

```
lamp(0, 200, 0)  
print("running")
```

step 4 · show a coloured lamp

```
def lamp(r, g, b):  
    on = f"\033[48;2;{r};{g};{b}m"  
    off = "\033[0m"  
    print(on + " " * 12 + off, end=" ")
```

```
lamp(0, 200, 0)  
print("running")
```

def gives a piece of code a name. r, g and b are its parameters.

step 4 · show a coloured lamp

```
def lamp(r, g, b):  
    on = f"\033[48;2;{r};{g};{b}m"  
    off = "\033[0m"  
    print(on + " " * 12 + off, end=" ")
```

```
lamp(0, 200, 0)  
print("running")
```

def gives a piece of code a name. r, g and b are its parameters.

you fill in the values when you call it: red, green, blue, each 0 to 255.

step 4 · show a coloured lamp

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def lamp(r, g, b):  
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```

```
lamp(0, 200, 0)  
print("running")
```

def gives a piece of code a name. r, g and b are its parameters.

you fill in the values when you call it: red, green, blue, each 0 to 255.

the strange codes inside? you do not have to understand them.

a function hides the details.

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you need to know **what** `lamp()` does, not **how**.

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`print()` and `input()` are functions too. did you miss their insides?

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next week, `led.set_color()` hides a whole device.

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you need to know **what** `lamp()` does, not **how**.

`print()` and `input()` are functions too. did you miss their insides?

next week, `led.set_color()` hides a whole device.

curious? open it up with your assistant.

where we are · next: step 5

✓ get the number of seconds from the user

`input()`

✓ check if the input is valid

`while`

✓ count down each second

`for`

✓ show a coloured lamp

`def`

5 decide which colour

`if / else`

step 5 · decide which colour

```
for remaining in range(seconds, 0, -1):  
    if remaining <= 5:  
        lamp(255, 200, 0)    # yellow  
    else:  
        lamp(0, 200, 0)      # green  
    print(remaining, "s")  
    time.sleep(1)  
  
lamp(255, 0, 0)              # red  
print("time is up!")
```


step 5 · decide which colour

```
for remaining in range(seconds, 0, -1):  
    if remaining <= 5:  
        lamp(255, 200, 0)    # yellow  
    else:  
        lamp(0, 200, 0)      # green  
    print(remaining, "s")  
    time.sleep(1)  
  
lamp(255, 0, 0)              # red  
print("time is up!")
```

if the condition is true, the first block runs. otherwise, the else block.

where we are · all five done

- ✓ get the number of seconds from the user `input()`
- ✓ check if the input is valid `while`
- ✓ count down each second `for`
- ✓ show a coloured lamp `def`
- ✓ decide which colour `if / else`

the whole program

```
import time

def lamp(r, g, b):
    on = f"\033[48;2;{r};{g};{b}m"
    off = "\033[0m"
    print(on + " " * 12 + off, end=" ")

text = input("How many seconds? ")
while not text.isdigit():
    text = input("Digits only, please: ")
```

the button copies all 23 lines. paste them into `timer.py` and read them there.

part 5

how our workshops work

⚙ stuck? ask in this order

1 your assistant

🌀 stuck? ask in this order

- 1 your assistant
- 2 your partner

🌀 stuck? ask in this order

- 1 your assistant
- 2 your partner
- 3 the team next to you

🎯 stuck? ask in this order

- 1 your assistant
- 2 your partner
- 3 the team next to you
- 4 nicolas

how to ask your assistant

say what you wanted to happen.
let it run the program, or send it a screenshot.
ask why, not only how.

↻ every week

type `/update-semester`: new material arrives in your folder.
your own work lives in my-code. everything else gets replaced.

where are you?

- ☐ OpenCode installed, course folder open
- ☐ assistant answers
- ☐ /onboarding done
- ☐ hello.py runs
- ☐ timer counts down
- ☐ timer shows its colours

until next monday:
form a team of two.
finish the setup and your timer.
read “the project” and “challenge 0”.
next week, your device. `lamp()`
becomes `led.set_color()`.

